pandas Basics: Selecting and Filtering

DATA SCIENCE BOOTCAMP

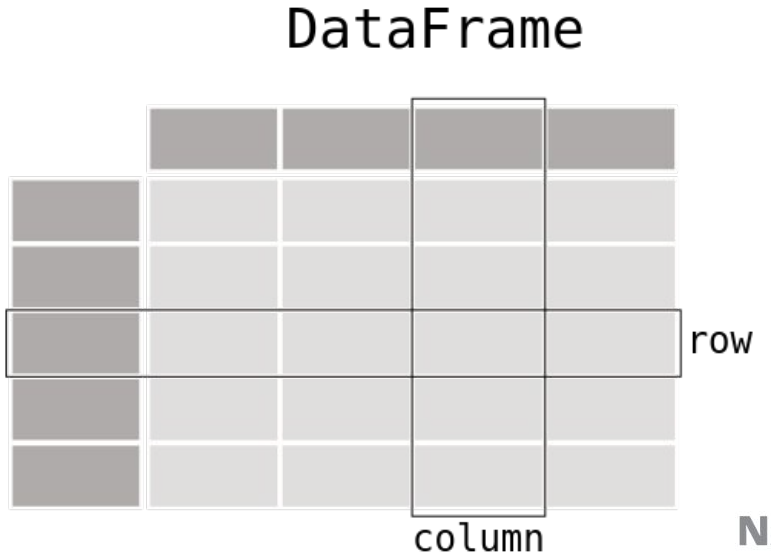
NASHVILLE SOFTWARE SCHOOL

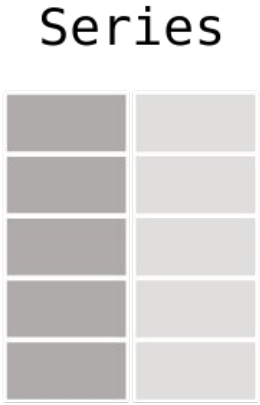
Objectives

1. Learn the difference between DataFrames and Series.
2. See how to use *.loc*, *.iloc*, and *[ ]* to subset and filter a DataFrame.

Dataframes and Series in pandas

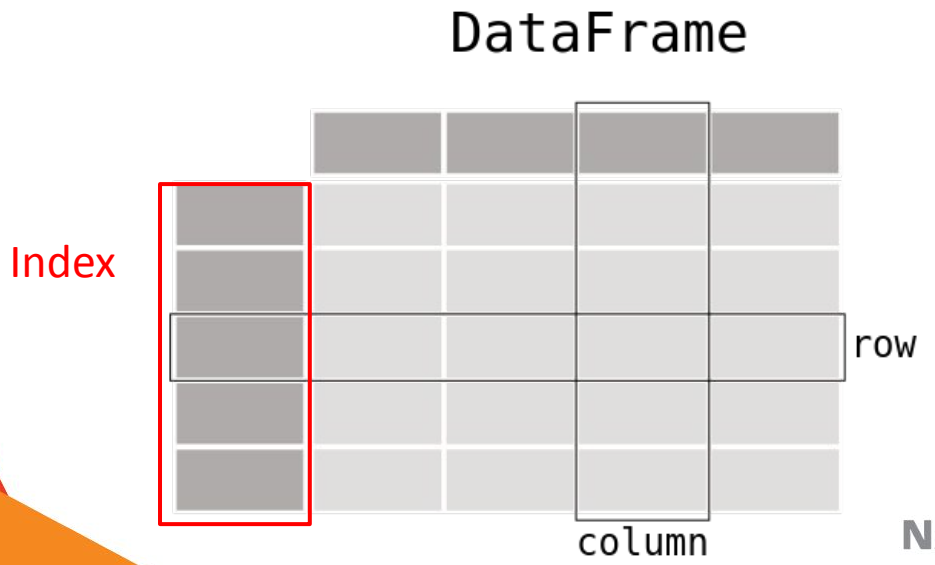
The primary tool for working with data in *pandas* is the **DataFrame**, an object which can hold tabular data.
You'll also often be working with *pandas* **Series**, a one-dimensional object. Single columns or rows of a DataFrame are Series.





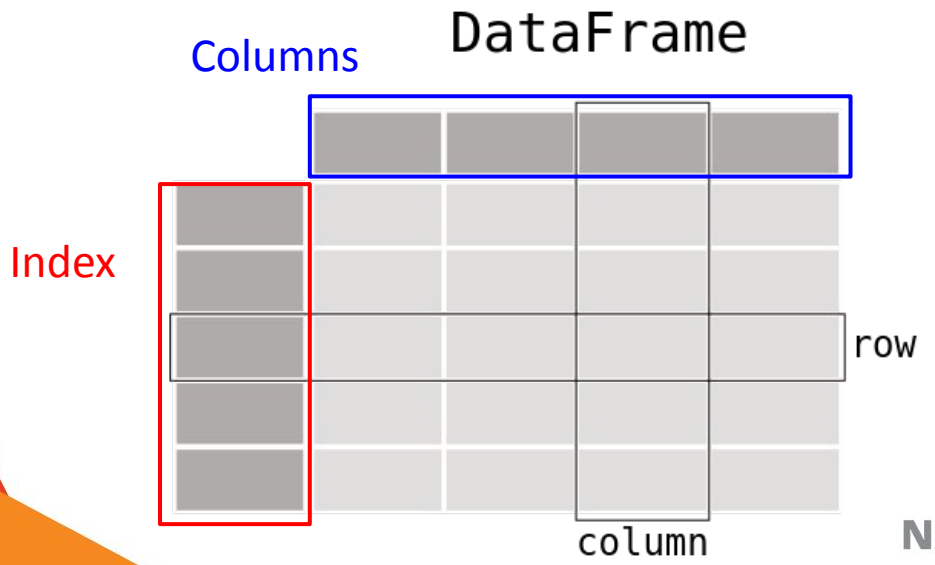
Dataframes and Series in pandas

Both DataFrames and Series have an **index**, which can be thought of as a row label.



Dataframes and Series in pandas

Both DataFrames and Series have an **index**, which can be thought of as a row label. DataFrames also have **columns**.



Dataframes and Series in pandas

```
1 platonic_solids
```

	Polyhedron	Vertices	Edges	Faces	Schläfli symbol	Vertex configuration
0	tetrahedron	4	6	4	{3, 3}	3.3.3
1	cube	8	12	6	{4, 3}	4.4.4
2	octahedron	6	12	8	{3, 4}	3.3.3.3
3	dodecahedron	20	30	12	{5, 3}	5.5.5
4	icosahedron	12	30	20	{3, 5}	3.3.3.3.3

Dataframes and Series in pandas

```
platonic_solids
```

	Polyhedron	Vertices	Edges	Faces	Schläfli symbol	Vertex configuration
0	tetrahedron	4	6	4	{3, 3}	3.3.3
1	cube	8	12	6	{4, 3}	4.4.4
2	octahedron	6	12	8	{3, 4}	3.3.3.3
3	dodecahedron	20	30	12	{5, 3}	5.5.5
4	icosahedron	12	30	20	{3, 5}	3.3.3.3.3

```
platonic_solids.index
```

```
RangeIndex(start=0, stop=5, step=1)
```

When reading from a csv, you'll get a RangeIndex by default, which just numbers the rows.

```
platonic_solids.columns
```

```
Index(['Polyhedron', 'Vertices', 'Edges', 'Faces', 'Schläfli symbol',
       'Vertex configuration'],
      dtype='object')
```

Selecting and Filtering

There are 3 major ways to select and filter:

1) df[ ]
   - Pass in either a column or list of columns.

```
platonic_solids[['Polyhedron', 'Edges']]
```

	Polyhedron	Edges
0	tetrahedron	6
1	cube	12
2	octahedron	12
3	dodecahedron	30
4	icosahedron	30

Selecting and Filtering

There are 3 major ways to select and filter:

1) df[ ]
   - Pass in either a column or list of columns in order to subset.

A single value will produce a Series.

```
1 platonic_solids['Polyhedron']
```

```
0     tetrahedron
1            cube
2      octahedron
3    dodecahedron
4     icosahedron
Name: Polyhedron, dtype: object
```

A list with a single value will produce a DataFrame.

```
1 platonic_solids[['Polyhedron']]
```

	Polyhedron
0	tetrahedron
1	cube
2	octahedron
3	dodecahedron
4	icosahedron

Selecting and Filtering

There are 3 major ways to select and filter:

1) df[ ]
   - Pass in either a column or list of columns in order to slice.
   - Pass in a Boolean Series in order to filter.

```
1 platonic_solids['Edges'] == 12
```

```
0    False
1     True
2     True
3    False
4    False
Name: Edges, dtype: bool
```

```
1 platonic_solids[platonic_solids['Edges'] == 12]
```

	Polyhedron	Vertices	Edges	Faces	Schläfli symbol	Vertex configuration
1	cube	8	12	6	{4, 3}	4.4.4
2	octahedron	6	12	8	{3, 4}	3.3.3.3

Selecting and Filtering

There are 3 major ways to select and filter:

1) df[ ]
   - When selecting based on multiple conditions, you can either do multiple .locs or use the & (or | for or) between conditions.

```
(
    platonic_solids
    .loc[platonic_solids['Vertices'] >= 8]
    .loc[platonic_solids['Faces'] < 15]
)
```

	Polyhedron	Vertices	Edges	Faces	Schläfli symbol	Vertex configuration
1	cube	8	12	6	{4, 3}	4.4.4
3	dodecahedron	20	30	12	{5, 3}	5.5.5

Selecting and Filtering

There are 3 major ways to select and filter:

1) df[ ]
   - When selecting based on multiple conditions, you can either do multiple .locs or use the & (or | for or) between conditions.

```
platonic_solids.loc[(platonic_solids['Vertices'] >= 8) & (platonic_solids['Faces'] < 15)]
```

	Polyhedron	Vertices	Edges	Faces	Schläfli symbol	Vertex configuration
1	cube	8	12	6	{4, 3}	4.4.4
3	dodecahedron	20	30	12	{5, 3}	5.5.5

Selecting and Filtering

There are 3 major ways to slice and filter:

2) df.loc[ ]
   - Label-based selection

```
platonic_solids.loc[3]
```

```
Polyhedron                dodecahedron
Vertices                            20
Edges                               30
Faces                               12
Schläfli symbol                 {5, 3}
Vertex configuration            5.5.5
Name: 3, dtype: object
```

Selecting and Filtering

```
platonic_solids_reindexed = platonic_solids.set_index('Polyhedron')
platonic_solids_reindexed
```

Polyhedron	Vertices	Edges	Faces	Schläfli symbol	Vertex configuration
tetrahedron	4	6	4	{3, 3}	3.3.3
cube	8	12	6	{4, 3}	4.4.4
octahedron	6	12	8	{3, 4}	3.3.3.3
dodecahedron	20	30	12	{5, 3}	5.5.5
icosahedron	12	30	20	{3, 5}	3.3.3.3.3

```
platonic_solids_reindexed.loc[['cube', 'icosahedron']]
```

Polyhedron	Vertices	Edges	Faces	Schläfli symbol	Vertex configuration
cube	8	12	6	{4, 3}	4.4.4
icosahedron	12	30	20	{3, 5}	3.3.3.3.3

Selecting and Filtering

There are 3 major ways to slice and filter:

2) df.loc[ ]
   - Label-based selection
   - Can select both columns and rows (df.loc[desired rows, desired columns]).
   - Note: When using .loc to select a range, it is inclusive of both ends of the range.

```
platonic_solids.loc[1:3, ['Polyhedron', 'Faces']]
```

	Polyhedron	Faces
1	cube	6
2	octahedron	8
3	dodecahedron	12

Selecting and Filtering

There are 3 major ways to slice and filter:

2) df.loc[ ]
   - Label-based selection
   - Can select both columns and rows (df.loc[desired rows, desired columns]).
   - Can also use a Boolean Series to select.

```
platonic_solids.loc[platonic_solids['Edges'] == 12, ['Polyhedron', 'Faces']]
```

	Polyhedron	Faces
1	cube	6
2	octahedron	8

Selecting and Filtering

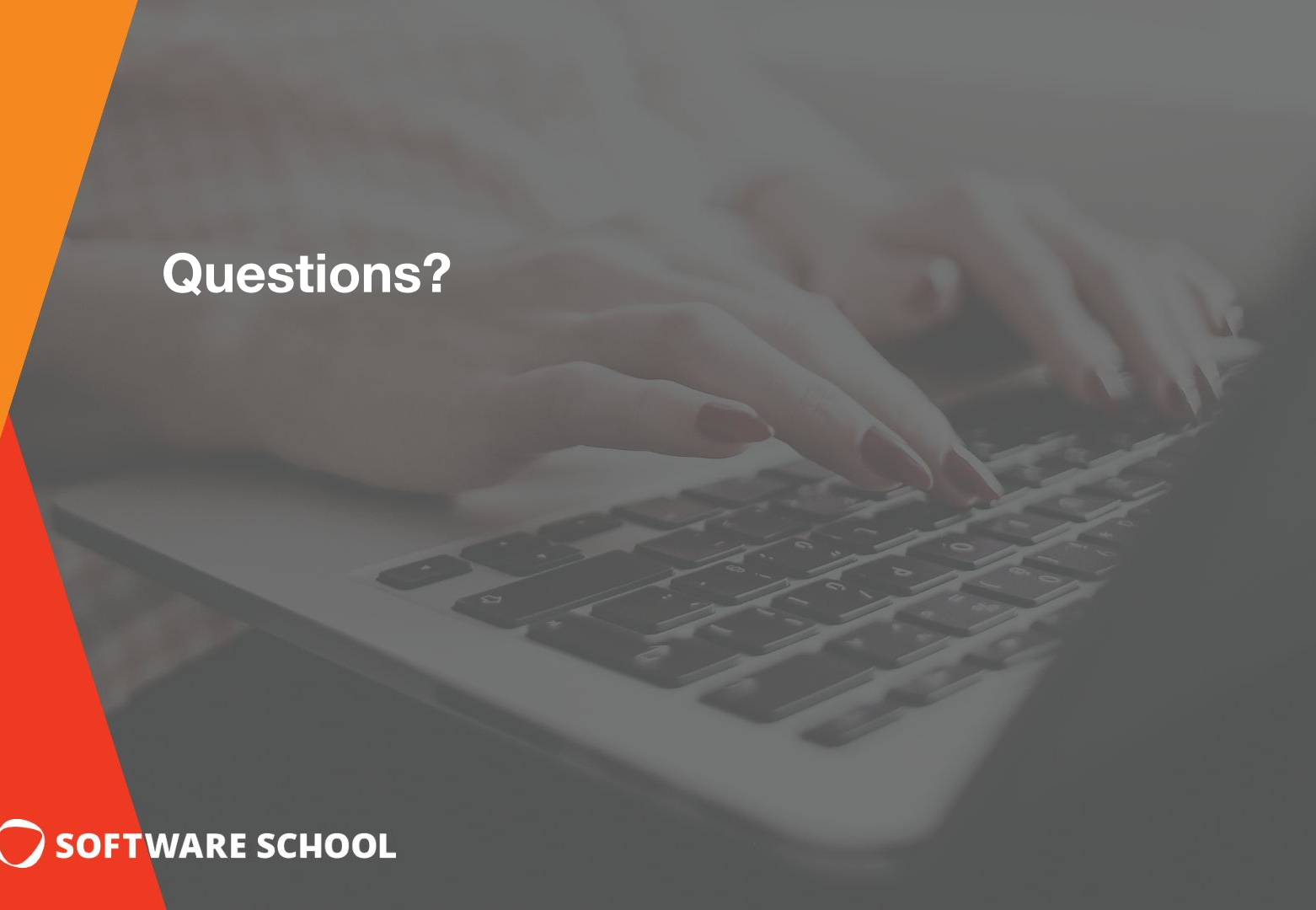
There are 3 major ways to slice and filter:

3) df.iloc[ ]
   - Integer-based selection
   - Can select both columns and rows (df.iloc[desired rows, desired columns]).
   - Note: Selecting a range with iloc includes the lower value but excludes the upper value.

```
platonic_solids.iloc[0:2, 0:3]
```

	Polyhedron	Vertices	Edges
0	tetrahedron	4	6
1	cube	8	12

Questions?



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